

One $\mathbb{Z}/2$ engine, two results that formally share it

the $\mathbb{Z}/2$ sign-averaging engine

$$\sum_{s \in \{\pm 1\}} s^k = 2 \cdot [k \text{ even}]$$

Lean: `sum_sign0f_pow_eq_{zero_of_odd, two_of_even}`

charpoly level

(elementary-symmetric)

$$\det(xI - A_s) = \mu_G$$

Godsil-Gutman gate: $\sigma^2 = 1$

`godsil_gutman` proven

*a shared atomic lemma
(grep-verifiable),
not a single meta-theorem*

moment level

(power-sum)

$$\text{tr}(A_s^d) = P_d$$

parity-closed walks gate: edges even

`sum_sign0f_prod_pow` kernel

parity gate: survives $\Leftrightarrow$ all even

`signAvg_ne_zero_iff`